

Mohit Kumar Rathod

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SKILLS

Languages: Python, SQL.

Backend Development: FastAPI, API Design, Microservices, Asynchronous Processing, Production Deployment

AI/LLMs: Large Language Models, RAG, Agentic AI Workflows, Conversational AI, LangGraph, LangChain, AI Orchestrations, Prompt Engineering, vLLM, Tool Calling

Cloud & DevOps: AWS EC2, ECS, ECR, S3, SQS, Lambda, Docker, Azure Databricks, Git, GitHub Actions

WORK EXPERIENCE

AI Engineer - Griphic

April 2025 - Present

- **Own the backend and AI infrastructure** for an AI-powered assessment platform supporting **1000+ parallel assessments at 99% uptime** across **5k+ daily evaluations**, leading architecture and implementation across assessment orchestration, LLM workflows, inference infrastructure, assessment evaluation, reporting, and observability.
- **Architected** an event-driven **AWS (Lambda, SQS, S3)** microservice to decouple compute-heavy report generation from synchronous backend APIs, slashing user API wait times from 45s to <200ms, eliminating timeout failures and delivering reports with 99.9% reliability.
- **Instrumented** end-to-end distributed tracing, metrics, and session execution context using **OpenTelemetry and the LGTM stack**, reducing production debugging time by **60%** and empowering cross-functional dev teams to diagnose complex multi-step LLM failures in minutes.
- Prototyped **real-time voice agents** using **Pipecat**, integrating **ElevenLabs TTS** with **Deepgram/Sarvam STT** and custom agents built on OpenAI enabling low-latency, streaming speech-to-speech communication.

Data Scientist - EarnIn

Aug 2024 - April 2025

- Trained and productionized **XGBoost models** on **10M+ samples and 400+ features** using PySpark, improving the existing algorithm's accuracy by **3%** and deploying scheduled inference on **Azure Databricks**.
- Reduced geospatial visualization latency from **60s to 10s** by migrating the existing visualization workflow to **KeplerGL**, improving responsiveness for geolocation analysis.

Data Scientist - Pegasystems (Internship)

Feb 2024 - Aug 2024

- Built an **Intelligent Document Processing** pipeline combining **Table Transformers, Donut, OCR, LLMs, and YOLOv8** to extract structured document and table data into CSV outputs with confidence scores.
- Developed **Streamlit-based interfaces** for GPT-powered document workflows, enabling interactive data processing and seamless integration between extracted content and LLM-based processing.

PROJECTS

Valerius - Multi-Model vLLM Inference Gateway — [GitHub](#)

June 2026 - Present

- Built a **FastAPI, vLLM, and Docker-based inference gateway** to provide a unified interface for locally hosted LLMs, supporting authenticated chat APIs, SSE token streaming, model discovery, and on-demand model switching.
- Implemented automatic **GPU/CPU backend selection** using NVIDIA runtime detection and CPU fallback configurations, enabling portable local deployments across heterogeneous environments.
- Productionized the gateway with an **OpenAI-compatible backend, Docker Compose, uv, Makefile automation, health checks, and smoke tests** for reproducible deployment and inference validation.

RAG-based Transactional Conversation System

June 2026

- Developed a production-style **FastAPI conversational backend** for transaction and merchant queries, implementing validated bulk ingestion, payload normalization, deterministic IDs, and modular API workflows.
- Built a **Qdrant-based semantic retrieval layer** with embeddings, metadata filtering, collection management, and indexed financial attributes to provide relevant transaction context for user queries.
- Integrated **Ollama/Llama and AgentField** into asynchronous RAG and agent workflows for contextual answer generation and source extraction, packaged as a **Dockerized multi-service application**.

EDUCATIONAL QUALIFICATIONS

Indraprastha Institute of Information Technology, Delhi

2022-2024

M.Tech, Computer Science & Engineering (AI)

CGPA: 8.08/10.0

Bhilai Institute of Technology, Durg

2018-2022

B.Tech (Hons.), Computer Science & Engineering

CPI: 9.03/10.0